# DATA 202 — Forum 3

Ch. 3 — "How We Know What a Number Means"

**Group members** (all three names):

1. _____ 2. _____ 3. _____

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*Week 8 — Friday class, after Fall Break*

Stone's argument: numbers don't speak for themselves — "more often than not, numbers are part of somebody's argument." To learn what a number means, ask Hans Rosling's question: "How would people have to live or what would people have to do to produce that number?" From a child mortality rate of 14 per 1,000, Rosling pictures Malaysian families with enough food, clean drinking water and mothers who can read. A French teacher knows that three 20-minute lessons are not two 30-minute ones, because school events keep cancelling the lessons. Authors dress their numbers in words and plots: the same migrant caravan was "trudging" or "surging," and Rosling's rising lines mean good news only because he tells us so. Measures of success carry their makers' values. The GDP counts a company that causes an oil spill as much as the company paid to clean it up, so "we tend to get what we measure so we should measure what we want." In Flint, Michigan, official cutoffs and words ("poisoned" or merely "exposed") shaped what the lead numbers meant, until residents and their allies used those numbers as witnesses against the officials who had hidden them.

In Weeks 6–7 we turned models into numbers: accuracy, precision, recall, cross-validation scores. This forum asks what those numbers mean.

*Discuss and write a response to each question below (~25 min).*

**1.** Rosling asks of every number, "How would people have to live or what would people have to do to produce that number?" and Stone adds that "a number is only the period at the end of a life story." In Week 7, a two-question tree on a blood sample was right for 87.5% of the test children: 42 of 48, all from one study. Do Rosling's exercise for that 87.5%: what did children, parents, clinics, researchers and our train/test split have to do to produce it? Then tell the story of the other 6 children: what would the tree's answer mean for them and their families?

**2.** Stone shows that the words around a number tell the story: the same migrant caravan was "trudging" in one report and "surging" in another. Our Week 6 spam filter caught 76% of the spam. Write two one-sentence headlines with that same number, one that makes the filter sound great and one that makes it sound bad. Which words did the work?

**3.** "The GDP doesn't care whether you produce goods or bads as long as you make money doing it." Accuracy is similar: it counts every right answer the same and every mistake the same. That is how a lazy "never spam" filter scored 88% while catching no spam at all, and a lazy "never Chinstrap" model scored 81% while finding no Chinstraps. Stone quotes an economist: "We tend to get what we measure so we should measure what we want." Pick one model from Weeks 6–7 (spam filter, Chinstrap finder, autism blood test). What would you want to measure instead of accuracy, or next to it, and whose values does your choice put into the number?

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**Submission.** Photograph or scan your annotated worksheet and upload it to Moodle by **Wednesday of the following week (Week 9)**. See Forum Instructions for the full process.